A Statistical Audit of LLM Calibration Heterogeneity Across Knowledge Domains

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STAT 215B, Spring 2026

What is calibration — and why does it matter?

LLMs are deployed across diverse domains — law, medicine, mathematics, ethics. Users treat model confidence as a reliability signal.

A model is **well-calibrated** if its stated confidence tracks empirical accuracy:

- 80% confident ⇒ correct 80% of the time

Miscalibration is typically reported as a **single number** (Expected Calibration Error):

$$\mathrm{ECE} = \sum_b \frac{|B_b|}{n} |\mathrm{acc}(B_b) - \mathrm{conf}(B_b)|$$

The problem: a single aggregate ECE hides *where* and *how much* miscalibration occurs across different domains.

Analogy: reporting one accuracy number across demographic groups is insufficient for a fairness audit. The same logic applies to calibration.

[45 sec]
Open with the definition — make sure the audience knows what calibration means before anything else. The formula is on the slide so you don't need to dwell on it; just say "ECE is the weighted average gap between confidence and accuracy across confidence bins."
The key move is the fairness analogy: just like we'd never accept a single accuracy number as a fairness audit, a single ECE number is not a calibration audit. Land that point before moving on — it's the whole motivation.
If anyone in the room works on ML deployment, you can add: "when a model says it's 90% confident, users treat that as a reliability signal — if that signal is only accurate on average, users in certain domains are being systematically misled."
[If asked: how does the ECE formula work?] You divide the confidence range $[0, 1]$ into equal-width bins. Every question is assigned to a bin based on its confidence value — e.g. a question where the model is 87% confident goes into the 0.8–0.9 bin. B_b is the set of *all* questions whose confidence falls in bin b, regardless of whether they were correct. $\mathrm{acc}(B_b)$ is the fraction of those questions the model got right. $\mathrm{conf}(B_b)$ is the simple average of confidence values for all questions in that bin. The comparison is: of all questions where the model said it was $\sim$85% confident, was it actually right 85% of the time? $|\mathrm{acc}(B_b) - \mathrm{conf}(B_b)|$ is the miscalibration in that bin, weighted by $|B_b|/n$ so larger bins contribute more.
Binning is based on whichever confidence measure you use. In our case that is the renormalized log-probability softmax. If you used verbal elicitation, you would bin on those verbal probabilities instead. The formula is agnostic to how confidence was obtained.

Why bin count matters: wider bins average out more variation in confidence within the bin, which can mask miscalibration. A model that is 81% confident on easy questions and 89% confident on hard ones in the same bin gets averaged to 85%, hiding within-bin misalignment. This is exactly why we ran the bin-count sensitivity check — to confirm subject rankings are stable across $b \in \{10, 15, 20, 25, 30\}$.

Research questions

Setting: MMLU benchmark — 14,042 questions, 57 subjects, 4 domains (STEM, Humanities, Social Sciences, Other) — queried with GPT-4o-mini and Llama-3-8B-Instruct.

1. **How much** of calibration variance is attributable to domain vs. subject vs. individual question?
2. **What question features** predict the direction and magnitude of miscalibration?
3. **Are patterns consistent** across models, or model-specific?

Framework

Three-level mixed model + isotonic calibration curves + empirical Bayes shrinkage + Benjamini–Hochberg FDR correction

[30 sec]
Read the three questions quickly — the audience will follow. For the framework block: “I won’t spend time on each method now; the key idea is that we want statistically rigorous answers, not just descriptive bar charts.”

Don’t explain what BH correction is here — that comes up naturally in the robustness slide. Just name-drop it so the audience knows it’s coming.

Data & confidence elicitation

MMLU: 14,042 four-choice questions, 57 subjects, 4 domains (questions → subjects → domains)

Two models:

- **GPT-4o-mini** (OpenAI API, `top_logprobs=20`)
- **Llama-3-8B-Instruct** (Colab T4, 4-bit quantization)

Confidence measure — renormalized softmax over the four answer tokens:

$$c_i = \max_{a \in \{A,B,C,D\}} \frac{e^{\ell_a}}{\sum_{a'} e^{\ell_{a'}}}$$

Primary outcome — signed calibration gap:

$$\mathrm{gap}_i = c_i - y_i \quad (y_i \in \{0,1\})$$

[45 sec]

Explain the hierarchy briefly: "MMLU has 57 subjects like clinical medicine, formal logic, high school math — grouped into 4 broad domains. That nested structure is what makes a multilevel model the right tool."

For the confidence measure: "We extract the log probability the model assigns to each of the four answer tokens, renormalize them to sum to 1, and take the max. This is a standard approach — it gives us a proper probability on the model's chosen answer."

For the signed gap: "The nice thing about using the signed gap instead of ECE is that it tells us the *direction* — positive means the model thinks it knows more than it does."

One methodological note if it comes up: we had to use top_logprobs=20 for GPT because with 5 logprobs, 55% of questions had at least one answer token missing, which would artificially inflate confidence estimates. That's a real data quality issue that took some debugging.

[Reference: gap vs. ECE — what is used where]

- **Calibration gap** (question-level, signed): outcome for the multilevel regression; BH t-tests test whether subject mean gap = 0; bar chart shows shrunken subject mean gaps (BLUPs); variance decomposition and ICC are computed on gap variance.
- **ECE** (subject-level scalar, unsigned): used to rank subjects (top-5 most miscalibrated); drives the reliability curves; cross-model Spearman ρ compares ECE rankings across models; sensitivity analysis checks ECE ranking stability across bin counts.
- The two are numerically nearly identical here because all subjects are uniformly overconfident — when every bin has conf > acc, ECE $\approx$ mean gap. But they serve different roles: gap supports signed inference, ECE is the standard scalar for calibration comparison.
- **Both models are fit separately.** The mixed model, BH tests, and BLUPs are run independently for GPT-4o-mini and Llama-3-8B-Instruct. Results are shown side-by-side. The cross-model slide then compares their ECE subject rankings via Spearman ρ.

[If asked: what exactly is the signed calibration gap?] The gap is computed at the *question level*: for each of the 14,042 questions, we take the model's renormalized confidence $c_i \in (0,1]$ (the max softmax probability over A/B/C/D) and subtract the binary correctness indicator $y_i \in \{0,1\}$. So $\mathrm{gap}_i = c_i - y_i$ is a number in $(-1,1)$ for every individual question. It is positive when the model was more confident than it deserved (overconfident) and negative when it was less confident than it deserved (underconfident).

From that question-level quantity we derive everything else:

- **Multilevel regression**: all 14,042 gap values are the outcome. The intercept (0.196) is the grand mean; the fixed effects (word count, entropy) are slopes; the random effects partition variance across subjects and domains.
- **Subject mean gap** (bar chart, Finding 3): simple average of gap_i across all questions in a subject. A subject with mean gap = 0.487 (moral scenarios) means that on average the model was 49 percentage points more confident than it was accurate.

Hierarchical model

Three-level mixed model

Questions (i) nested in subjects (j) nested in domains (k):

$$\mathrm{gap}_{ijk} = \mu + \mathbf{x}_{ijk}^{\top}\boldsymbol{\beta} + u_k + v_{jk} + \varepsilon_{ijk}$$

Fixed effects

μ: global intercept (mean overconfidence)

$\boldsymbol{\beta}$: question-level covariates (word count, entropy, negation, max choice length) — all standardized

Random effects

$u_k \sim \mathcal{N}(0,\ \sigma^2_{\mathrm{domain}})$

$v_{jk} \sim \mathcal{N}(0,\ \sigma^2_{\mathrm{subject}})$

$\varepsilon_{ijk} \sim \mathcal{N}(0,\ \sigma^2_{\varepsilon})$

Estimated via REML (`statsmodels.MixedLM`)

ICC decomposition: what fraction of calibration gap variance is between-subject vs. within-subject?

[30 sec]

Walk through the model briefly: “We model the signed calibration gap — confidence minus accuracy — at the question level, with random intercepts for both subject and domain.”

Emphasize the key question: “The ICC decomposition will tell us whether the miscalibration is driven by specific subjects, or whether it’s spread evenly across all questions regardless of subject.”

Mention REML: “We use restricted maximum likelihood, which gives unbiased variance component estimates — important here since the variance decomposition is our main inferential target.”

[If asked: why REML instead of standard MLE?] Both maximize a likelihood over the variance components, but they differ in *what* likelihood they maximize.

MLE treats the fixed effect estimates $\hat{\mu}, \hat{\boldsymbol{\beta}}$ as known with certainty when computing residuals — as if no degrees of freedom were spent estimating them. This causes MLE to systematically *underestimate* variance components, analogous to dividing by n instead of $n - p$ when computing a sample variance.

REML marginalizes out the fixed effects first and maximizes the likelihood of a set of contrasts orthogonal to the fixed effect space. This accounts for the uncertainty in $\hat{\mu}$ and $\hat{\boldsymbol{\beta}}$, effectively dividing by $n - p$ rather than n, and yields unbiased variance component estimates.

Why it matters here: $\sigma^2_{\mathrm{subject}}$ and the ICC are our primary inferential quantities, not nuisance parameters on the way to fixed effect p-values. MLE would push the ICC toward zero, understating the degree to which subject membership explains calibration variation. REML is the standard choice whenever variance decomposition is the inferential goal.

[If asked: what does each term in the formula represent?]

$$\mathrm{gap}_{ijk} = \underbrace{\mu}_{\text{global mean}} + \underbrace{\mathbf{x}_{ijk}^{\top}\boldsymbol{\beta}}_{\text{question features}} + \underbrace{u_k}_{\text{domain shift}} + \underbrace{v_{jk}}_{\text{subject shift}} + \underbrace{\varepsilon_{ijk}}_{\text{residual}}$$

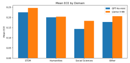
- $\mu = 0.196$: the grand mean calibration gap — every question, on average, has confidence 0.196 higher than accuracy. This is the global overconfidence from RLHF fine-tuning.
- $\mathbf{x}_{ijk}^{\top}\boldsymbol{\beta}$: the contribution of question-level features (word count, entropy, negation, max choice length) to that question’s gap above or below

Finding 1: heterogeneity is real but concentrated at the question level

RQ1: How much of calibration variance is attributable to domain vs. subject vs. question?

Table: Variance decomposition (both models)

Level	GPT-4o-mini Var	GPT-4o-mini ICC	Llama-3-8B Var	Llama-3-8B ICC
Domain	0.000	0.000	0.000	0.000
Subject	0.013	**0.079**	0.004	**0.020**
Residual	0.150	0.922	0.190	0.980



Domain-level mean ECE (descriptive)

Both models: between-subject structure exists but is modest.

GPT: **7.9%** between-subject ICC.
Llama: **2.0%** — more diffuse miscalibration.

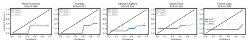
[30 sec]
Lead with the structure: "Both models show that the vast majority of calibration variance — 92 to 98% — is at the question level. But a meaningful slice is systematically concentrated at the subject level. That's not noise — it's structure."
Highlight the difference: "GPT shows 7.9% between-subject ICC, Llama shows 2.0%. Llama's miscalibration is more diffuse — spread more evenly across subjects rather than concentrated in particular domains."
[Important: what ICC does and does not say] ICC = Intraclass Correlation Coefficient = $\sigma^2_{\text{subject}}/\text{total variance}$. Intuition: if you drew two questions at random from the *same* subject, ICC is how correlated their calibration gaps would be due to shared subject membership. ICC = 0 means subject tells you nothing; ICC = 1 means every question in a subject has the same gap.
ICC = 7.9% does NOT mean calibration barely varies across subjects. It means 7.9% of total question-level variance is attributable to subject membership. That 7.9% is still large enough to produce a $13\times$ range in mean subject gaps (0.037 to 0.487) — the two facts are compatible. Most question-to-question variability is noise regardless of subject (the 92%), but the systematic component that does exist is substantively meaningful.
Be upfront about domain ICC = 0: "Domain-level variance is estimated at zero. That doesn't mean domains don't matter descriptively — we see clear differences in the bar chart. But the model can't separate domain-level variance from subject-level variance because the subject random effects absorb whatever between-domain structure exists. With only 4 domain groups we also have very low power to detect a non-zero domain component even if one were present."
[If asked: what is the ICC?] ICC stands for Intraclass Correlation Coefficient. It answers: *of all the variance in the calibration gap, what fraction is explained by knowing which group a question belongs to?*
Formally, for the subject level:

$$\text{ICC}_{\text{subject}} = \frac{\sigma^2_{\text{subject}}}{\sigma^2_{\text{domain}} + \sigma^2_{\text{subject}} + \sigma^2_{\text{residual}}}$$

A useful intuition: if you drew two questions at random from the *same* subject, their calibration gaps would be correlated — that shared component is exactly $\text{ICC}_{\text{subject}}$. If ICC were 0, knowing the subject tells you nothing about a question's gap. If ICC were 1, every question in a subject would have the same gap. At 7.9%, subject membership explains a modest but real fraction — enough to produce a $13\times$ range in mean subject gaps.
[If asked: how do we get the variance components and ICC?] Fitting the mixed model via REML directly outputs the variance components σ^2_{domain}, $\sigma^2_{\text{subject}}$, and $\sigma^2_{\text{residual}}$ as part of the optimization — no separate step needed.

Finding 2: all 57 subjects are overconfident on average

RQ1: Visual evidence of subject-level miscalibration pattern



The 5 most miscalibrated subjects shown. All 57 subjects have positive mean calibration gap. In the high-confidence region where data is concentrated, curves lie below the diagonal throughout. Isotonic (solid) and kernel-smoothed (dashed) curves agree where support exists; divergence at low confidence reflects empty bins.

[30 sec]

Lead with the universal claim carefully: "All 57 subjects have a positive mean calibration gap — they are overconfident on average. These 5 curves show what that looks like in the region where data is actually concentrated, which for most subjects is the high-confidence end."

Be precise: the below-diagonal pattern holds in the region of support (roughly 0.85–1.0 for most subjects). It is NOT correct to say every point on every curve is below the diagonal — 23/57 subjects have parts of their isotonic curve above the diagonal, but this is at low-confidence bins where there are essentially no data points. The mean gap being positive is the correct universal claim.

On the formal_logic kernel curve specifically: the dashed curve reads near 1.0 from confidence 0.03 to 0.53. This is a smoothing artifact — formal_logic has zero questions in those bins (126 total questions, 89 of which are in the 0.95–1.00 bin). The kernel is smoothing over empty space and should not be interpreted there. The isotonic curve is more honest: it just holds the last observed value flat across empty bins.

Point out isotonic vs. kernel agreement where it matters: "In the region where data exists — above 0.7 or so — the solid and dashed lines track each other, which validates the monotonicity assumption in that range."

Transition to Finding 3: "The next slide shows the full picture across all 57 — the key result is not just that they are all overconfident on average, but that the magnitude varies 13-fold."

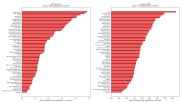
[If asked: what does '5 most miscalibrated' mean — ECE or signed gap?] These are the 5 subjects with the highest ECE, computed per subject as the weighted mean absolute deviation between the reliability curve and the diagonal across 20 bins. ECE is always positive — it measures magnitude but not direction. Since all 57 subjects have positive mean gap (universally overconfident), the top 5 by ECE and top 5 by mean gap are essentially the same ranking here. But they are conceptually different: if some subjects were underconfident, ECE would still flag them as miscalibrated while the signed gap would show negative values.

[If asked: what does below vs. above the diagonal mean?] Below the diagonal = overconfident: the model says 0.9 but is right only 0.5 of the time. Users trust it more than they should. Above the diagonal = underconfident: the model says 0.6 but is right 0.8 of the time. It hedges more than necessary. We find universal overconfidence — all curves below the diagonal — which is consistent with RLHF fine-tuning pushing probability mass toward the chosen answer to produce confident, helpful-sounding responses. Base models typically show better calibration kadavath2022language, nakkiran2025trained. Overconfidence is the worse deployment failure: an underconfident model prompts users to verify; an overconfident model in legal or medical settings actively misleads users who treat confidence as a reliability signal.

[Limitation: support is concentrated at high confidence] The confidence distribution is extremely right-skewed — the median confidence across all

Finding 3: universal overconfidence, 13× range in magnitude

RQ1: How large and heterogeneous is subject-level miscalibration?



57/57 subjects rejected after BH correction ($\alpha = 0.05$).

Subject mean gap > 0: overconfident on average; < 0: underconfident on average.

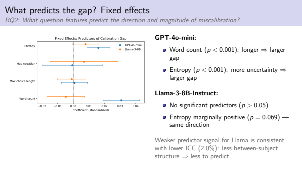
But the *magnitude* varies drastically (GPT-4o-mini):

- **Max**: moral scenarios (mean gap $= 0.487$)
- **Min**: marketing (mean gap $= 0.037$)
- **Range**: 13×

Humanities subjects cluster at the top; STEM and "Other" at the

[45 sec]
"57/57 rejected — but don't read too much into the count. RLHF fine-tuning pushes every subject above zero by default: the global intercept is 0.196, meaning there's a baseline overconfidence floor before any subject-level differences enter. With 100–1,500 questions per subject and that strong global signal, the t-tests have enough power to flag everything. BH has 57 near-zero p-values and nothing to correct.
The interesting finding isn't that all 57 reject — it's that the *magnitude* varies 13-fold. Some subjects are essentially well-calibrated; others are off by nearly 50 percentage points. That heterogeneity is invisible in a single aggregate number."
[If asked: where does empirical Bayes / shrinkage come in?] The bars in this chart are *shrunken* subject mean gaps — BLUPs (Best Linear Unbiased Predictors) from the mixed model, not raw subject averages. Subjects have very different sample sizes (100 to 1,534 questions). A small subject with 100 questions has much noisier estimates — it could look extremely miscalibrated just by chance. Empirical Bayes shrinkage corrects for this by pulling each subject's estimate toward the grand mean (0.196), with more shrinkage for smaller subjects and less for larger ones.
The BLUP $\hat{v}_{jk}$ is the subject random effect from the REML fit. It is already a shrunken estimate by construction — REML estimates random effects by shrinking them toward zero (i.e., toward the grand mean), with the degree of shrinkage determined by the ICC and sample size. So the shrunken subject mean is simply $\mu + \hat{v}_{jk}$: grand mean plus the shrunken subject deviation. This is equivalent to a weighted average of the subject's own observed mean and the grand mean — you get it for free from the model fit, no separate step needed.
What is the 13×? Simply the ratio of the largest to smallest mean gap: moral scenarios (0.487) divided by marketing (0.037) $\approx 13\times$. The model is 13 times more overconfident on moral scenarios than on marketing. That heterogeneity is invisible in any single aggregate ECE number.
"A gap of 0.487 means the model is, on average, nearly 50 percentage points more confident than it is accurate. A gap of 0.037 means it is essentially well-calibrated. Those are qualitatively different deployment situations."
[If asked: why is 57/57 not a meaningful finding?] Intuition: imagine testing whether a coin is slightly biased. With 10 flips you might not detect it. With 1,000,000 flips you would detect even a 50.001% bias as "statistically significant" — but that doesn't mean it matters. Same here: with 14,042 questions and a model already globally overconfident by 0.196 on average, you have enough power to flag even a trivially small true effect. Even marketing (gap $= 0.037$, the best subject) gets rejected because the sample is large enough to detect anything nonzero.
[If asked: exactly how does BH work here — what values are computed?] For each of the 57 subjects we run a **one-sample *t*-test** on the question-level calibration gaps, testing H_0: mean gap $= 0$ (the subject is well-calibrated on average). This gives 57 p-values, one per subject.
BH then:

1. Sort the p-values in ascending order: $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(57)}$.
2. For each rank k, compute the BH threshold $k/57 \times \alpha$



[30 sec]

Note: these are **two separate models**, each fit independently on its own data. GPT and Llama each produce their own β estimates — the two columns on this slide are not from a single joint model.

Lead with the GPT entropy finding — it's counterintuitive: "When GPT-4o-mini spreads its probability mass across options, expressing uncertainty, the gap actually gets larger. The model isn't reliably correct more often when it's hedging; it's miscalibrated even in its expression of uncertainty."

Then contrast with Llama: "Llama shows no significant predictors. The trend for entropy is in the same direction but doesn't reach significance. This is consistent with Llama having less between-subject structure overall — there's simply less systematic variance to explain."

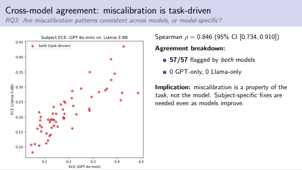
[If asked: why do GPT and Llama differ in fixed effect significance?] GPT's ICC is 7.9% — its miscalibration is organized by subject, concentrating in subjects like abstract algebra and moral scenarios. These subjects also tend to have longer, more ambiguous questions. So word count and entropy are picking up real signal because they correlate with the hard subjects. Llama's ICC is only 2.0% — its miscalibration is more diffuse, not concentrated in particular subjects. There is no such structure for question features to exploit. On top of that, Llama's residual variance is larger in absolute terms ($\sigma^2_\varepsilon = 0.190$ vs. 0.150 for GPT), so the same fixed effects face more noise, inflating standard errors and making significance harder to reach.

[If pushed on the fixed effects interpretation — important limitation] The question-level predictors (word count, entropy) are standardized across the full dataset, not within subjects. This means the fixed effect estimates blend two things that should ideally be separated:

- **Within-subject effect**: within abstract algebra, do longer questions get larger gaps?
- **Between-subject effect**: do subjects whose questions tend to be longer (like abstract algebra) also tend to have larger gaps?

Subject random intercepts absorb the between-subject *baseline* in the gap, but they do not remove the between-subject covariation between the predictors and the gap. The clean fix is **group-mean centering**: subtract each subject's mean word count from each question's word count before fitting. That isolates the pure within-subject effect. Without it, the fixed effects are dataset-level associations, not clean question-level effects.

The honest answer if pressed: "The fixed effects should be interpreted as associations between question features and the calibration gap, not as cleanly isolated question-level effects. This is a real limitation of RQ2. The primary findings — ICC decomposition, the $13\times$ magnitude range, and the cross-model $\rho = 0.846$ — do not depend on the fixed effects at all and are unaffected by this."



[45 sec]

[Big picture: how this slide connects to earlier findings] Three separate claims have been made so far — it is easy to conflate them:

1. **Universal overconfidence** (global intercept = 0.196): all 57 subjects are overconfident because the model itself is globally biased by RLHF. The 57/57 BH rejection is mostly detecting this global effect, not between-subject differences.
2. **Between-subject heterogeneity in magnitude** (ICC = 7.9%, $13\times$ range): on top of the global bias, subjects vary in how badly miscalibrated they are. 7.9% of total variance is between-subject — modest, but enough to produce a $13\times$ range in mean gaps that matters practically.
3. **Task-driven miscalibration** (this slide, $\rho = 0.846$): whatever between-subject variation exists is consistent across two very different models. This answers a different question than ICC — not "how much variation is there?" but "where does that variation come from?"

The cross-model result does not contradict the low ICC. It says: the subject-level structure that does exist (7.9%) is driven by the task, not the model.

[What does Spearman $\rho = 0.846$ exactly say?] Spearman rank correlation ranks all 57 subjects by ECE separately for each model, then measures how well those two orderings agree. $\rho = 0.846$ means the rankings are strongly correlated — subjects GPT ranks as most miscalibrated, Llama also tends to rank near the top, and vice versa. It is a rank correlation (not Pearson), so it is robust to outliers and non-linear scaling differences between the two models — it only asks "do the orderings agree?", not "are the absolute ECE values the same?" $\rho = 1.0$ would be perfect rank agreement; $\rho = 0$ would mean no relationship. 0.846 is strong. The bootstrap 95% CI $[0.734, 0.910]$ confirms this is not a fluke driven by a few extreme subjects.

What does it imply? If miscalibration were model-specific — a quirk of GPT's RLHF or Llama's supervised fine-tuning — you would expect low correlation: each model would have its own idiosyncratic hard subjects. The high ρ means the hardness is a property of the knowledge domain itself. Combined with 57/57 overlap, there is strong evidence that no subject escapes miscalibration and that the ranking of which subjects are worst is determined by the task, not the model.

This is a strong result — emphasize it: "Both a large proprietary API model (GPT-4o-mini) and a small open-weights model running at 4-bit quantization (Llama-3-8B-Instruct) produce essentially the same pattern. Despite the difference in scale, architecture, and deployment context, the Spearman correlation of 0.846 means subject rankings are highly consistent."

The key takeaway: "This is not a GPT problem or a Llama problem. It's a knowledge domain problem. The subjects that are hard to calibrate on are hard for both models we tested. That means recalibration needs to happen at the subject level, not the model level."

Point to the scatter plot: "Every dot is red — both models reject. The dots above the diagonal are subjects where Llama is worse; below are where GPT is worse. But both are bad."

[If asked: what can and cannot be concluded from $\rho = 0.846$?] Can conclude:

- The subject rankings are highly consistent across models — what GPT finds hardest, Llama also finds hardest.

Sensitivity analysis: are findings robust?

Three stress tests — run on both models.

1. ECE binning & NLCS

$b \in \{10, 15, 20, 25, 30\}$ bins.

GPT: $\rho \geq 0.997$
Llama: $\rho \geq 0.984$

vs. unbinned NLCS: $\rho \geq 0.861$

Rankings stable.

2. FDR threshold

BH at $\alpha \in \{0.01, 0.05, 0.10\}$.

GPT: 56/57/57 rejected
Llama: 55/57/57 rejected

Not driven by α.

3. Model structure

3-level vs. 2-level model.

GPT: $\mathrm{ICC}_{\mathrm{subj}} = 0.079$
Llama: $\mathrm{ICC}_{\mathrm{subj}} = 0.020$
(identical in both specs)

ICC robust.

[45 sec]
Frame briefly: "Before trusting these results, I stress-tested the three main modeling choices across both models."
Column 1: "We vary the number of confidence bins used to compute ECE — 10, 15, 20, 25, 30 — and check whether subject rankings change. To be clear: ECE bins questions by their *confidence value*, not by the gap. For each bin, we compare mean accuracy to mean confidence across the questions in that bin. Changing bin count shifts absolute ECE values slightly but barely moves the subject ordering: $\rho \geq 0.997$ for GPT. We also compare against the unbinned NLCS as a completely bin-free alternative: $\rho \geq 0.861$. Our rankings aren't artifacts of binning choices."
Column 2: "At alpha=0.01, GPT still rejects 56/57 and Llama rejects 55/57. The two Llama subjects that drop out at the strictest threshold have adjusted p-values of 0.014 and 0.031 — just above the cutoff. The finding is not alpha-sensitive."
[If asked: what does varying the BH α tell us?] BH α controls the expected proportion of false discoveries among all rejected tests. At $\alpha = 0.05$, at most 5% of rejections are expected to be false; at $\alpha = 0.01$, at most 1%.
If rejections were borderline — p-values clustered just below the threshold — tightening α would flip many decisions. The fact that results barely move (57/57 $\rightarrow$ 56/57 for GPT, 57/57 $\rightarrow$ 55/57 for Llama) shows almost all subjects have adjusted p-values so far below any reasonable threshold that the BH correction barely matters. The two or three subjects that drop at $\alpha = 0.01$ have adjusted p-values of 0.014 and 0.031 — just above the cutoff, not strongly non-significant.
Someone cannot dismiss the finding by saying "you chose a lenient α" — at the strictest threshold tested you still reject 55–56 of 57 subjects.
This also reinforces the earlier point that 57/57 is not a deep finding: p-values are near zero for most subjects because the global intercept (0.196) is large relative to per-subject standard errors. BH has almost nothing to correct because the evidence is overwhelming regardless of α. The sensitivity check confirms robustness, but the interesting result remains the $13\times$ magnitude range, not the binary rejection count.
Column 3: "The two-level and three-level models give identical ICC estimates — but be honest about why: domain variance is already estimated at zero in the three-level model, so dropping the domain level is numerically a no-op. This is not an independent validation. The useful interpretation is narrower: the three-level model is not artificially inflating subject-level ICC by misattributing domain variance to subjects. Since domain variance = 0, the subject ICC of 7.9% (GPT) is not contaminated by a spurious domain effect. The stronger robustness evidence is columns 1 and 2."

[If asked: why Spearman specifically for the bin sensitivity check?] Our conclusions rest on subject *rankings*, not absolute ECE values: the top-5 miscalibrated subjects are selected by rank, and the cross-model comparison ($\rho = 0.846$) is itself a rank correlation. When we change bin count, Takeaways

1. **The unit of analysis is the subject, not the model.** Aggregate ECE hides a 13× range in miscalibration magnitude. 57/57 subjects rejected is the wrong headline — the magnitude variation is what matters.
2. **Miscalibration is task-driven, so subject-level fixes are needed.** Spearman $\rho = 0.846$ across two very different models means the same subjects are hardest regardless of architecture or training. Global temperature scaling cannot fix subject-specific structure.
3. **The audit framework generalizes.** Multilevel model + BH correction + shrinkage applies to any benchmark with a hierarchical structure — a reusable tool for honest calibration evaluation.
4. **Open questions remain.** Two models is suggestive, not definitive. We can describe *which* subjects are hard but not *why* — abstractness, training data rarity, or something else is still unknown.

Next steps: subject-specific temperature scaling; broader cross-model comparison; extend to open-ended generation.

[45 sec]

Don't read the bullets — synthesize across all four:

"The core finding is that the subject is the right unit of analysis for calibration, not the model. Aggregate ECE hides a 13-fold range; 57/57 rejected sounds impressive but is mostly a consequence of large sample size and global overconfidence — the magnitude variation is the finding that matters."

"The reason subject-level recalibration is needed, not model-level, is that $\rho = 0.846$ across two very different models tells us the hardness is in the knowledge domain itself. You cannot train your way out of it."

"We're honest about what we don't know: two models is suggestive not definitive, and we can describe which subjects are hard but not why. That's the natural next step."

Land the practical implication last: "If you are deploying an instruction-tuned model on professional tasks — legal, medical, mathematical — the aggregate calibration number is not what you should be reporting. The gap is 13 times worse on some subjects than others, and that pattern holds across models."

End confidently: "The framework is the contribution. It gives you a statistically honest answer about where any model is unreliable, using only token-level logprobs and standard multilevel modeling."

Leave 30 sec for questions.